

RAM is the first place the CPU looks into for data before it tries to load anything from the storage media, which are not-so-surprisingly also known as secondary memory.

RAM holds the data in “active use” by processes running on the system and the OS itself. Like everything else that makes up a PC, you will be spoiled by choices when it comes to RAM modules. But first, let’s see what makes up a good RAM module.

Key points of RAM

What are the two most important things to look for in the RAM when we read about them? DDR rating and size. It is so because they are the most important ones affecting the performance of a PC. In case you were unaware, DDR stands for “Double Data Rate” indicating that they can transfer at twice the speed. You may ask “twice the speed of what?” And the answer would

	
Max Turbo Frequency ?	3.06 GHz
Cache ?	8 MB SmartCache
Bus Speed ?	4.8 GT/s QPI
# of QPI Links ?	1
TDP ?	130 W
VID Voltage Range ?	0.800V-1.375V
Supplemental Information	
Embedded Options Available ?	No
Datasheet	View now
Memory Specifications	
Max Memory Size (dependent on memory type) ?	24 GB
Memory Types ?	DDR3 800/1066
Max # of Memory Channels ?	3
Max Memory Bandwidth ?	25.6 GB/s
Physical Address Extensions ?	36-bit
ECC Memory Supported ?	No

CPUs also determine the max amount of memory that they can address